

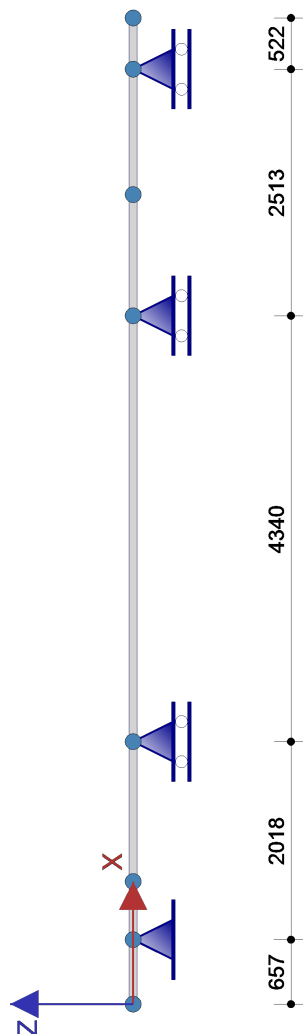
File :.....71_constructie_advies\2846 5.1 Vloer nieuw-2.xbe2

Table of contents

1.1 NODES.....	1
1.2 BEAMS.....	2
1.3 PROFILES.....	2
1.4 LOAD CASES.....	2
1.5 LOAD CASE 1 Dead load Including self-weight.....	3
1.6 LOAD CASE 2 Live load.....	4
2.1 LOAD CASES.....	5
2.1.1 Node displacements.....	20
2.1.2 Envelope node displacements.....	20
2.1.3 Reaction forces.....	22
2.1.4 Envelope reaction forces.....	22
2.1.5 Beam forces.....	22
2.1.6 Envelope beam forces.....	23
2.1.7 Beam stresses.....	24
2.2 ULTIMATE LIMIT STATES (ULS).....	24
2.2.2 Envelope reaction forces.....	26
2.2.3 Envelope beam forces.....	28
2.3 SERVICE LIMIT STATES (SLS).....	28
2.3.2 Envelope node displacements.....	30
2.4 REINFORCEMENT.....	32
2.4.1 Longitudinal reinforcement.....	32
2.4.2 Longitudinal reinforcement - Ultimate limit state (ULS).....	32
2.4.3 Longitudinal reinforcement - Service limit state (SLS).....	33
2.4.4 Stirrups.....	33
2.4.5 Shear reinforcement - Ultimate limit state (ULS).....	33

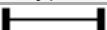
Applied standards : NEN-EN 1992-1-1+C1:2011/NB:2016+A1:2020 nl

Consequence class : CC1

Gravity acceleration g : 9,81 m/s²**1 Input Data****1.1 NODES**

Node Number	Coordinates		Restrains		
	X [mm]	Z [mm]	Tx	Tz	Ry
1	0	0			
2	1250	0			
3	8250	0			
4	10050	0			
5	657	0	A	A	
6	2675	0		A	
7	7015	0		A	
8	9528	0		A	

1.2 BEAMS

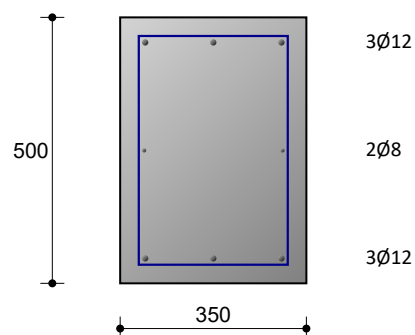
Beam Number	Node		Beam type	Profile	Length [mm]
	from	to			
1	1	2		Profile 2	1250

Beam Number	Node		Beam type	Profile	Length [mm]
	from	to			
2	2	3		Profile 2	7000
3	3	4		Profile 2	1800

1.3 PROFILES

Profile Number	Name	Weight [kg/m]	E [N/mm ²]	A [mm ²]	I _y [mm ⁴]	Wy;el_1 [mm ³]	Wy;el_2 [mm ³]
1	Profile 2	437,5	6748	1,75E5	3,6458E9	1,4583E7	1,4583E7

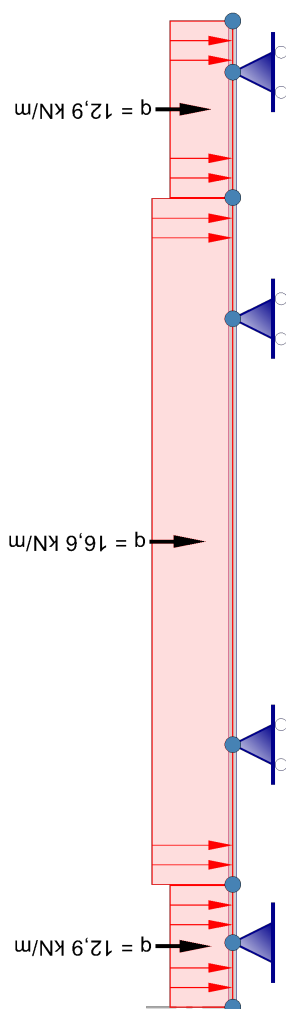
Profile 2



Element type	Beam	Structural Class S4
Prefab	no	
Concrete grade	C20/25	Creep coefficient 2,70
Steel grade	B500B	Granule diameter 31,50 mm
Number of stirrup sections	2	Angle compression strut 40
	<u>Top side</u>	<u>Bottom side</u>
Env. class	XC4	XC4
Concrete surface	Can be checked	Can be checked
ΔCdev	5 mm	
Cover	35 mm	35 mm EN 1992-1-1 (4.1)
Nominal cover c _{nom}	35 mm	35 mm
Side reinforcement in calculation	no	

1.4 LOAD CASES

no.	Description	Type	ψ ₀	ψ ₁	ψ ₂
1	Dead load	Dead load incl. self-weight	1,00	1,00	1,00
2	Live load	A:domestic	0,40	0,50	0,30

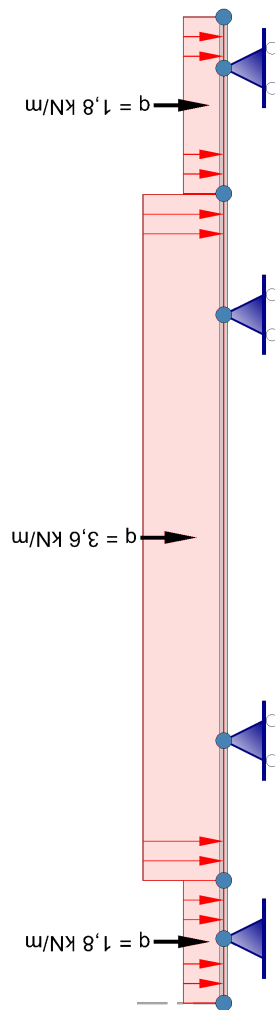
1.5 LOAD CASE 1 Dead load Including self-weight

*) Loads due to self-weight are not drawn!

Total self-weight: : 4313 kg.

1.5.1 Beam loads

Type	Loads			Distance from		
	q1	q2	Angle	Node	a [mm]	L [mm]
q	-4,292 kN/m	-4,292 kN/m	0,0	1	0	1250
q	-16,600 kN/m	-16,600 kN/m	0,0	1	1250	6990
q	-12,900 kN/m	-12,900 kN/m	0,0	1	0	1240
q	-12,900 kN/m	-12,900 kN/m	0,0	1	8250	1800
q	-4,292 kN/m	-4,292 kN/m	0,0	2	0	7000
q	-4,292 kN/m	-4,292 kN/m	0,0	3	0	1800

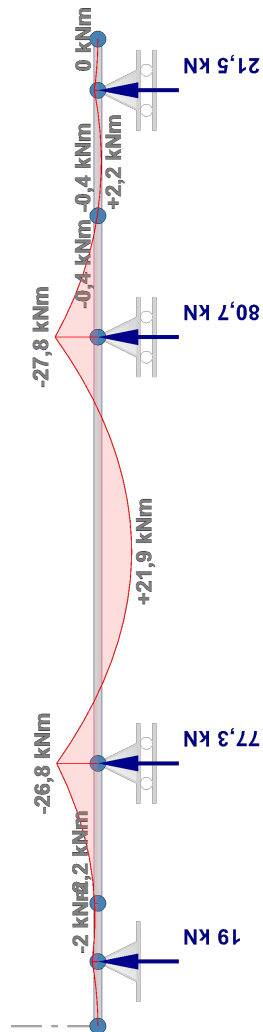
1.6 LOAD CASE 2 Live load**1.6.1 Beam loads**

Type	Loads			Distance from		
	q1	q2	Angle	Node	a [mm]	L [mm]
q	-1,800 kN/m	-1,800 kN/m	0,0	1	8250	1800
q	-1,800 kN/m	-1,800 kN/m	0,0	1	0	1250
q	-3,600 kN/m	-3,600 kN/m	0,0	1	1250	6990

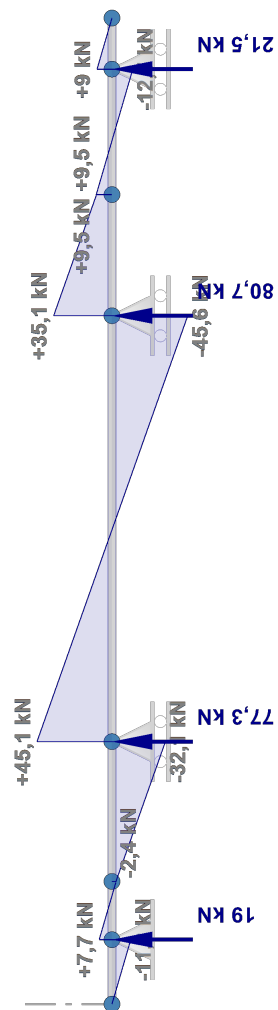
2 Calculation Results

2.1 LOAD CASES

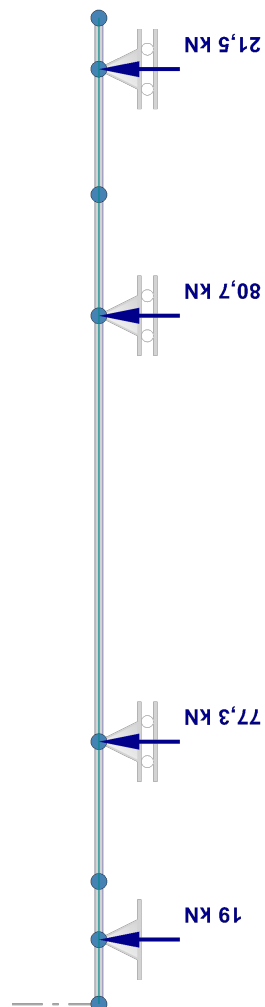
Geometric linear analysis



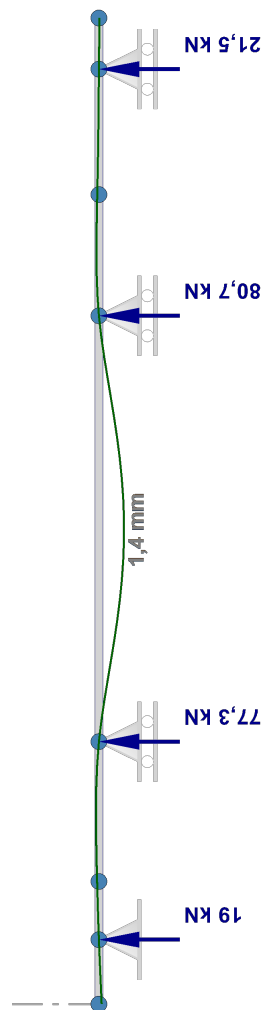
M-line - 1 Dead load



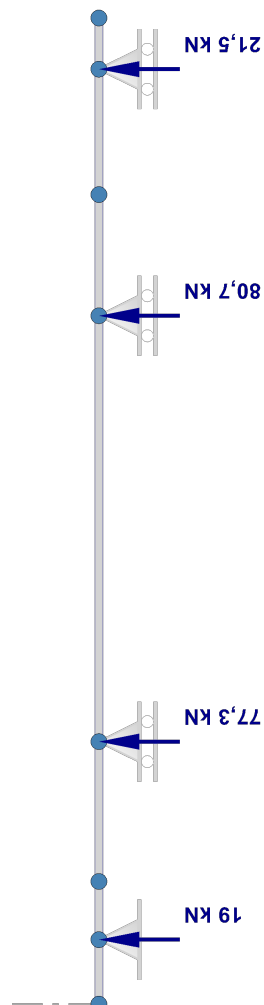
D-line - 1 Dead load



N-line - 1 Dead load



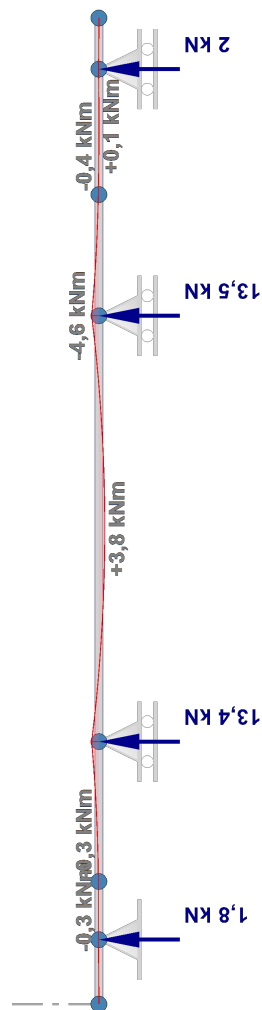
Displacements - 1 Dead load



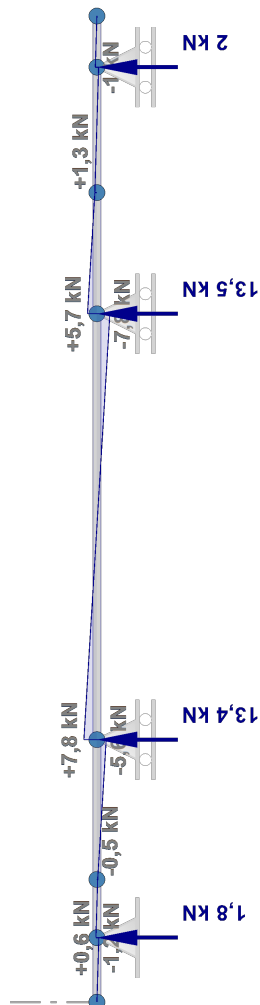
Soil pressure - 1 Dead load

d

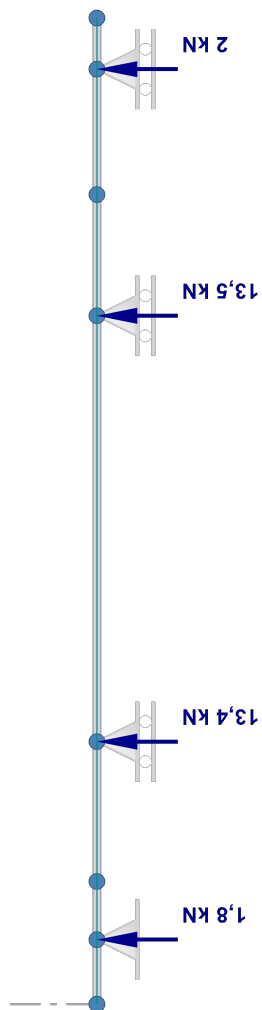
i



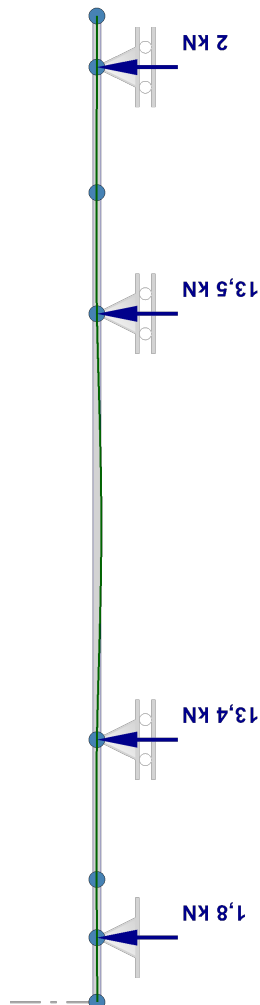
M-line - 2 Live load



D-line - 2 Live load

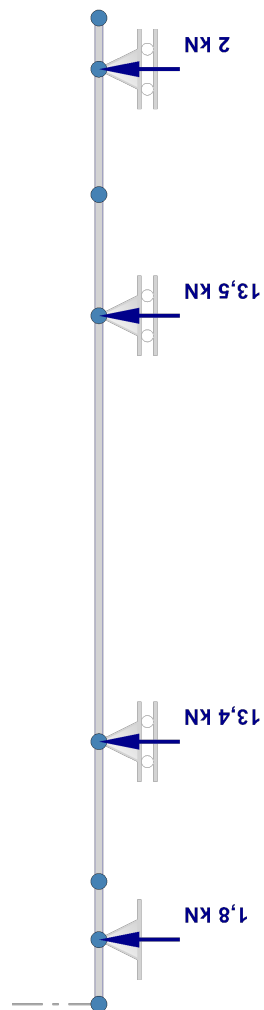


N-line - 2 Live load



Displacements - 2 Live load

d



2.1.1 Node displacements

Node Number	Load case	dx [mm]
1	1	0,0
	2	0,0
2	1	0,0
	2	0,0
3	1	0,0
	2	0,0
4	1	0,0
	2	0,0
5	1	0,0
	2	0,0
6	1	0,0
	2	0,0
7	1	0,0
	2	0,0
8	1	0,0
	2	0,0
Minimum / maximum v		
1	1	0,0
1	1	0,0
1	1	
2	1	
6	1	
7	1	

2.1.2 Envelope node displacements

Node Number	Load case	dx [mm]
1	1	0,0
	2	0,0
2	1	0,0
	2	0,0
3	1	0,0
	2	0,0
4	1	0,0
	2	0,0
5	1	0,0
	2	0,0
6	1	0,0
	2	0,0
7	1	0,0
	2	0,0
8	1	0,0
	2	0,0
Minimum / maximum v		
1	1	0,0
1	1	0,0
1	1	

Soil pressure - 2 Live load

dz [mm]	dr [mrad]
-0,1	0,2
0,0	0,0
0,1	0,1
0,0	0,0
0,1	-0,1
0,0	0,0
0,0	-0,1
0,0	0,0
0,0	0,2
0,0	0,0
0,0	-0,5
0,0	-0,1
0,0	0,5
0,0	0,1
0,0	-0,1
0,0	0,0
values	
-0,1	
0,1	
	-0,5
	0,5

dz [mm]	dr [mrad]
-0,1	0,2
0,0	0,0
0,1	0,1
0,0	0,0
0,1	-0,1
0,0	0,0
0,0	-0,1
0,0	0,0
0,0	0,2
0,0	0,0
0,0	-0,5
0,0	-0,1
0,0	0,5
0,0	0,1
0,0	-0,1
0,0	0,0
values	
-0,1	

Node Number	Load case	dx [mm]	dz [mm]	dr [mrad]
2	1		0,1	
6	1			-0,5
7	1			0,5

2.1.3 Reaction forces

Node Number	Load case	Fx [kN]	Fz [kN]	My [kNm]
5	1		18,984	
	2		1,782	
6	1		77,251	
	2		13,417	
7	1		80,654	
	2		13,476	
8	1		21,494	
	2		1,979	
Minimum / maximum values				
5	2		1,782	
7	1		80,654	

2.1.4 Envelope reaction forces

Node Number	Load case	Fx [kN]	Fz [kN]	My [kNm]
5	1		18,984	
	2		1,782	
6	1		77,251	
	2		13,417	
7	1		80,654	
	2		13,476	
8	1		21,494	
	2		1,979	
Minimum / maximum values				
5	2		1,782	
7	1		80,654	

2.1.5 Beam forces

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	1		0,000	0,000	0,000
		5		0,000	-11,295	-3,710
		5		0,000	7,689	-3,710
			1104	0,000	0,000	-1,991
		2		0,000	2,377	-2,173
	2	1		0,000	0,000	0,000
		5		0,000	-1,183	-0,388
		5		0,000	0,599	-0,388
			990	0,000	0,000	-0,289

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1		2		0,000	0,468	-0,350
2	1	2		0,000	-2,377	2,173
		6		0,000	-32,147	-26,771
		6		0,000	45,103	-26,771
			3584	0,000	0,000	21,914
		7		0,000	-45,568	-27,781
		7		0,000	35,086	-27,781
		3		0,000	-9,451	-0,381
	2	2		0,000	-0,468	0,350
		6		0,000	-5,598	-4,672
		6		0,000	7,818	-4,672
			3597	0,000	0,000	3,818
		7		0,000	-7,806	-4,644
		7		0,000	5,671	-4,644
		3		0,000	-1,261	-0,386
3	1	3		0,000	9,451	0,381
			550	0,000	0,000	2,217
		8		0,000	-12,520	-2,342
		8		0,000	8,974	-2,342
		4		0,000	0,000	0,000
	2	3		0,000	1,261	0,386
			700	0,000	0,000	0,055
		8		0,000	-1,040	-0,245
		8		0,000	0,940	-0,245
		4		0,000	0,000	0,000

2.1.6 Envelope beam forces

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	5		0,000	-11,295	-3,710
	1	5		0,000	7,689	-3,710
	2	5		0,000	-1,183	-0,388
	2		990	0,000	0,000	-0,289
	2	2		0,000	0,468	-0,350
2	1	2		0,000	-2,377	2,173
	1	6		0,000	45,103	-26,771
	1		3584	0,000	0,000	21,914
	1	7		0,000	-45,568	-27,781
	1	7		0,000	35,086	-27,781
	1	3		0,000	-9,451	-0,381
3	1	3		0,000	9,451	0,381
	2	3		0,000	1,261	0,386
	1		550	0,000	0,000	2,217
	1	8		0,000	-12,520	-2,342
	1	8		0,000	8,974	-2,342

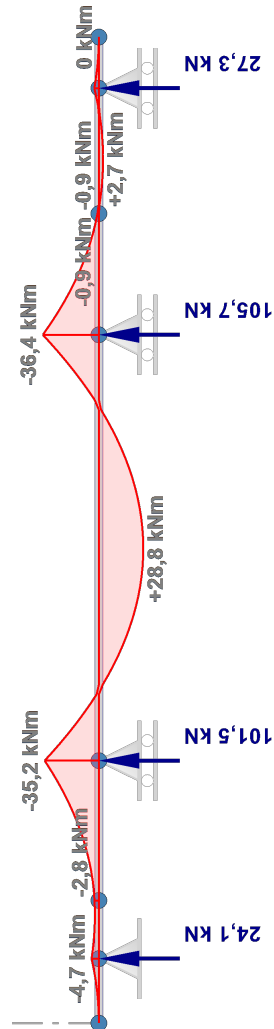
2.1.7 Beam stresses

Beam Number	Load case	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]	sigma1 [N/mm ²]	sigma2 [N/mm ²]
1	1	1	0	0,000	0,000	0,000	0,0	0,0
			0	0,000	-0,001	0,000	0,0	0,0
			657	0,000	7,689	-3,710	0,3	-0,3
1	2	1	0	0,000	0,000	0,000	0,0	0,0
			0	0,000	-0,001	0,000	0,0	0,0
			657	0,000	0,599	-0,388	0,0	0,0
2	1	2	0	0,000	-2,377	-2,173	0,1	-0,1
			1425	0,000	45,103	-26,772	1,8	-1,8
2	2	2	0	0,000	-0,468	-0,350	0,0	0,0
			1425	0,000	7,818	-4,673	0,3	-0,3
3	1	3	0	0,000	9,451	-0,381	0,0	0,0
			550	0,000	0,000	2,217	-0,2	0,2
			1278	0,000	8,974	-2,342	0,2	-0,2
3	2	3	0	0,000	1,261	-0,386	0,0	0,0
			700	0,000	0,000	0,055	0,0	0,0
			1278	0,000	0,940	-0,245	0,0	0,0

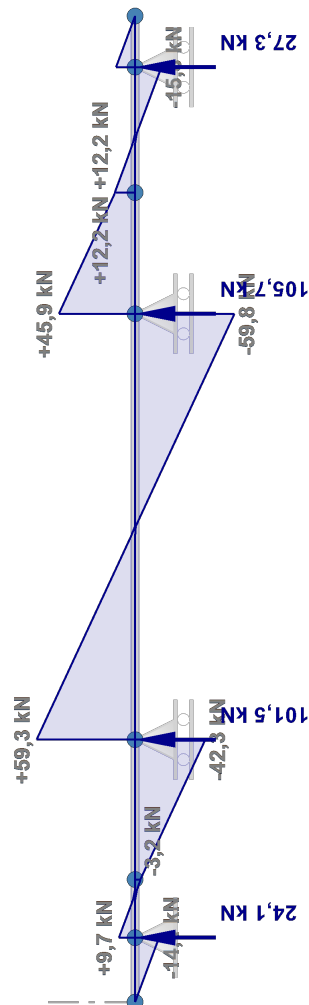
2.2 ULTIMATE LIMIT STATES (ULS)**2.2.1 Load combinations****Geometric linear analysis**

Combination Number	Description	Type
1	ULS(6.10a)	ULS
2	ULS(6.10b)	ULS

Combination Number	Load case ($\psi \times \gamma$)				
	1	2			
1	1,00x1,22	0,40x1,35			
2	1,00x1,08	1,00x1,35			



Envelop M-line



Envelop D-line

2.2.2 Envelope reaction forces

Node Number	Combination Number	Fx [kN]	Fz [kN]	My [kNm]
5	1		24,123	
	2		22,908	
6	1		101,491	
	2		101,543	
7	1		105,675	
	2		105,300	
8	1		27,292	
	2		25,886	
Minimum / maximum values				
5	2		22,908	
7	1		105,675	

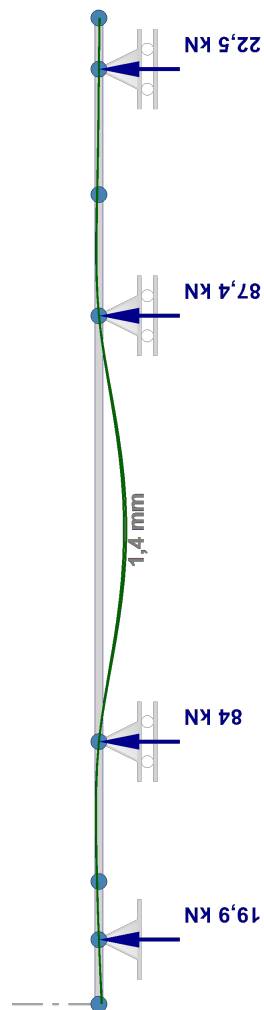
2.2.3 Envelope beam forces

Beam Number	Combination Number	Node Number	x-local [mm]	Nx-local [kN]	Vz-local [kN]	My-local [kNm]
1	1	5		0,000	-14,418	-4,736
	1	5		0,000	9,704	-4,736
	2	5		0,000	-13,795	-4,532
	2		1091	0,000	0,000	-2,554
	2	2		0,000	3,199	-2,819
2	1	2		0,000	-3,153	2,840
	1	7		0,000	-59,808	-36,400
	1	7		0,000	45,867	-36,400
	2	6		0,000	59,266	-35,221
	2		3586	0,000	0,000	28,820
3	1	3		0,000	-12,211	-0,673
	2	3		0,000	11,909	0,933
	1		556	0,000	0,000	2,724
	1	8		0,000	-15,836	-2,990
	1	8		0,000	11,456	-2,990

2.3 SERVICE LIMIT STATES (SLS)**2.3.1 Load combinations****Geometric linear analysis**

Combination Number	Description	Type
3	SLS Permanent	SLS Permanent
4	SLS Quasi permanent	SLS Quasi permanent
5	SLS	SLS

Combination Number	Load case (ψ x γ)				
	1	2			
3	1,00x1,00				
4	1,00x1,00	0,30x1,00			
5	1,00x1,00	0,50x1,00			



2.3.2 Envelope node displacements

Node Number	Combination Number	dx [mm]
1	3	0,0
	5	0,0
2	3	0,0
	5	0,0
3	3	0,0
	5	0,0
4	3	0,0
	5	0,0
5	3	0,0
	5	0,0
6	3	0,0
	5	0,0
7	3	0,0
	5	0,0
8	3	0,0
	5	0,0
Minimum / maximum v		
1	3	0,0
1	3	0,0
1	5	
2	5	
6	5	
7	5	

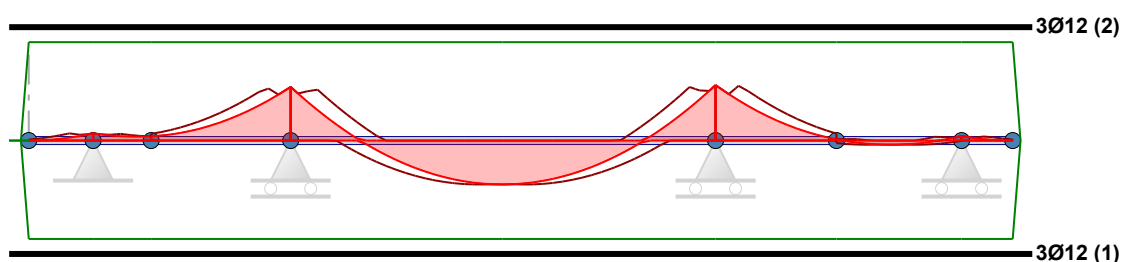
Envelop displacements

dz [mm]	dr [mrad]
-0,1	0,2
-0,2	0,2
0,1	0,1
0,1	0,1
0,1	-0,1
0,1	-0,1
0,0	-0,1
0,0	-0,1
0,0	0,2
0,0	0,2
0,0	-0,5
0,0	-0,5
0,0	0,5
0,0	0,5
0,0	-0,1
0,0	-0,1
values	
-0,2	
0,1	
	-0,5
	0,5

2.4 REINFORCEMENT

2.4.1 Longitudinal reinforcement

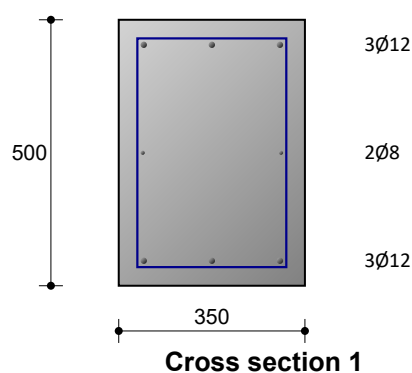
no.	from x [mm]	to x [mm]	Length [mm]	Side	Reinforcement	z [mm]	Ld begin [mm]	Ld end [mm]	Weight [kg]
1	-200	10250	10450	Bottom	3Ø12	-459	120	120	27,8
2	-200	10250	10450	Top	3Ø12	-41	120	120	27,8
3	-100	10150	10250	Bottom	2Ø8	-250	100	100	8,1
Total									63,8

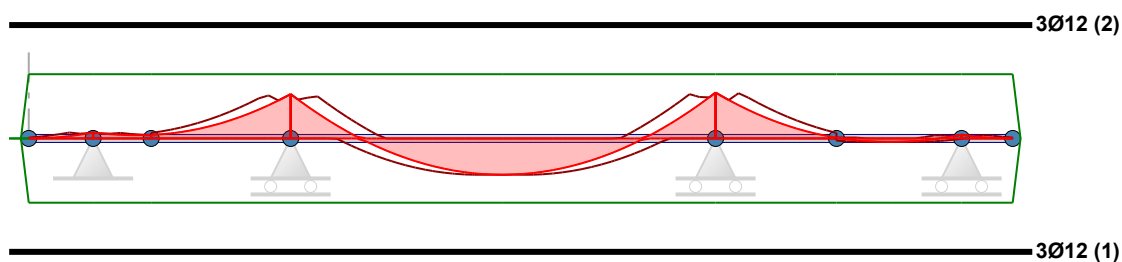


Envelope shifted M-line (ULS)

2.4.2 Longitudinal reinforcement - Ultimate limit state (ULS)

x [mm]	Sect.	MEd [kNm]	MRd [kNm]	xu [mm]	xu,max [mm]	Side	Reinforcement	Remarks
0	1	-0,7	-64,5	45,1	228,3	Top	3Ø12	
657	1	-4,7	-64,5	45,1	228,3	Top	3Ø12	
7015	1	-36,4	-64,5	45,1	228,3	Top	3Ø12	
9528	1	-3,0	-64,5	45,1	228,3	Top	3Ø12	
4836	1	28,8	64,5	45,1	228,3	Bottom	3Ø12	
8806	1	2,7	64,5	45,1	228,3	Bottom	3Ø12	
9528	1	0,2	64,5	45,1	228,3	Bottom	3Ø12	





Envelope shifted M-line (SLS)

2.4.3 Longitudinal reinforcement - Service limit state (SLS)

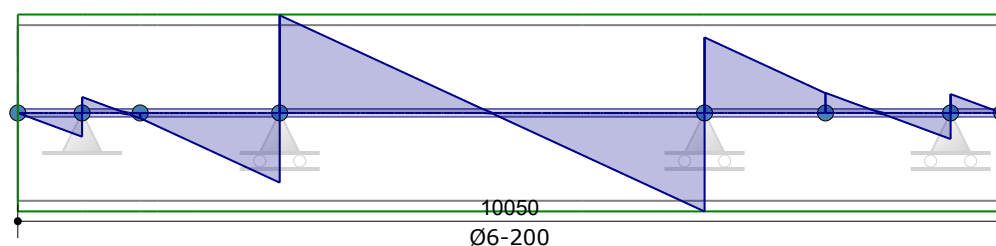
Control of cracking without direct calculation

...EN 1992-1-1 art.7.3.3

x [mm]	Sect.	Mk [kNm]	MRk [kNm]	s [mm]	s,max [mm]	Ø [mm]	Ø,max [mm]	Remarks
0	1	-0,5	-42,1	128,0	128,0	12,0	9,0	
657	1	-3,9	-42,1	128,0	128,0	12,0	9,0	
7015	1	-30,1	-42,1	128,0	128,0	12,0	9,0	
9528	1	-2,5	-42,1	128,0	128,0	12,0	9,0	
4835	1	23,8	42,1	128,0	128,0	12,0	9,0	
8807	1	2,2	42,1	128,0	128,0	12,0	9,0	
9528	1	0,2	42,1	128,0	128,0	12,0	9,0	

2.4.4 Stirrups

from x [mm]	to x [mm]	Length [mm]	Stirrups
0	10050	10050	Ø6-200 (2s.)



Envelope D-line (ULS)

2.4.5 Shear reinforcement - Ultimate limit state (ULS)

...EN 1992-1-1 art.6.2.1

x [mm]	VEd [kN]	VRd,c [kN]	VRd,s [kN]	VRd,max [kN]	Side	Stirrups	Remarks
0	0,0	-53,3	-59,7	-562,1	Top	Ø6-200 (2s.)	
657	0,0	-53,3	-59,7	-562,1	Top	Ø6-200 (2s.)	
1250	-3,2	-53,3	-59,7	-562,1	Top	Ø6-200 (2s.)	
1278	-4,0	-53,3	-59,7	-562,1	Top	Ø6-200 (2s.)	

x [mm]	VEd [kN]	VRd,c [kN]	VRd,s [kN]	VRd,max [kN]	Side	Stirrups	Remarks
1425	-8,0	-53,3	-59,7	-562,1	Top	Ø6-200 (2s.)	
1800	-18,3	-53,3	-59,7	-562,1	Top	Ø6-200 (2s.)	
5765	-25,5	-53,3	-59,7	-562,1	Top	Ø6-200 (2s.)	
7000	-13,5	-53,3	-59,7	-562,1	Top	Ø6-200 (2s.)	
657	9,7	53,3	59,7	562,1	Bottom	Ø6-200 (2s.)	
7000	35,2	53,3	59,7	562,1	Bottom	Ø6-200 (2s.)	
10050	0,0	53,3	59,7	562,1	Bottom	Ø6-200 (2s.)	